INTRODUCTION

Rapid population growth and continuous urbanization have created a global environmental crisis that threatens planetary sustainability. Approximately 75% of global greenhouse gas emissions originate from energy production, while around 80% of global energy needs are still met by fossil fuels [1]. This excessive dependence on fossil energy not only threatens global climate stability but also creates air pollution that endangers public health. The transportation sector, as the world's second-largest energy consumer utilizing approximately 30% of total global energy annually, plays a crucial role in CO_2 emissions contribution [2]. These negative impacts are further exacerbated by the rapid growth of conventional motor vehicles in developing countries, which exponentially adds to the emission burden. The high level of pollution has significantly damaged air quality and the environment, making the transformation towards sustainable transportation systems an urgent necessity to achieve net-zero emission targets and protect future generations [3].

Electric vehicles have emerged as a key technological solution for decarbonizing the transportation sector that significantly contributes to global emissions. Research demonstrates that electric vehicles can reduce greenhouse gas emissions by 30%–35% compared to conventional vehicles throughout their lifecycle, with greater reduction potential as electricity grids transition toward renewable energy sources [4], [5]. Global adoption of electric vehicles has experienced rapid growth, with sales reaching over 18% of total new vehicle sales in 2023, up from only 2% in 2018 [6]. The capability of electric vehicles to integrate renewable energy sources such as photovoltaics and energy storage systems not only reduces dependence on fossil fuels but also creates a sustainable and environmentally friendly transportation ecosystem [7], [8]. This transformation is supported by continuous innovations in battery technology and charging infrastructure development to address range and charging time challenges.

Despite the promising potential of electric vehicles, several significant barriers continue to hinder widespread adoption among consumers. Comprehensive literature reviews identify high upfront purchase costs, limited charging infrastructure availability, and range anxiety as the most frequently cited obstacles to EV adoption. High initial costs remain a critical barrier, as EVs are generally more expensive than comparable internal combustion engine vehicles, primarily due to battery technology expenses. Range anxiety, defined as the fear of running out of battery power before reaching a charging station, affects a significant portion of potential EV buyers and creates psychological barriers to adoption [9], [10], [11]. Additionally, inadequate charging infrastructure presents logistical challenges, with research showing that only 34% of prospective buyers are willing to wait more than 30 min for charging, while most existing charging stations require lengthy charging periods [12]. These interconnected barriers create a cascading effect that significantly impacts consumer confidence and delays the transition to sustainable transportation systems.

Among these barriers, inadequate charging infrastructure remains one of the most critical bottlenecks requiring strategic planning and careful site selection. The placement of EVCS involves a multidimensional decision problem that must account for interconnected factors such as electrical grid capacity, land availability, traffic flow patterns, user accessibility, and integration with existing urban infrastructure [13], [14], [15]. Empirical studies indicate that poorly located charging stations often experience low utilization rates, leading to economic inefficiencies and spatial inequities in service accessibility [16]. Many existing EVCS siting approaches rely on simplified criteria or single objective optimization methods that are insufficient to capture complex spatial interactions and diverse stakeholder requirements in urban contexts [17], [18]. As a result, fragmented planning practices continue to limit the effectiveness of electrification strategies and slow the transition toward more sustainable urban mobility systems [19].

A review of recent EVCS site selection studies indicates that disaster resilience has received relatively limited attention within prevailing infrastructure planning methodologies. Most existing studies employ conventional MCDM approaches that prioritize economic, technical, social, and environmental criteria without explicitly integrating disaster related considerations [20], [21], [22], [23]. At the same time, several GIS based fuzzy MCDM applications in cities such as Ankara, Istanbul, and Beijing demonstrate the effectiveness of hybrid methods for identifying suitable EVCS locations based on factors such as economic feasibility, environmental impact, and accessibility [20], [21], [24]. The Fuzzy AHP method has been shown to effectively address uncertainty and subjectivity in expert driven decision processes, making it appropriate for complex infrastructure planning problems [25], [26]. Nevertheless, these studies are largely situated within conventional urban planning contexts and generally do not operate disaster risk or emergency preparedness as explicit decision layers in EVCS site selection.

This methodological limitation becomes more evident when considering the needs of disaster-prone regions where infrastructure resilience is a central planning concern. Existing literature provides limited examples of EVCS site selection frameworks that systematically integrate disaster risk indicators within MCDM based spatial analysis, despite the recognized importance of resilient infrastructure in hazard exposed environments [27], [28]. This gap is particularly relevant for developing regions located in seismically active zones such as the Pacific Ring of Fire, where transportation infrastructure must simultaneously support routine mobility and emergency response during extreme events. Addressing this contextual challenge requires an application focused integration of disaster related spatial layers into established decision support frameworks rather than the development of entirely new methodological paradigms.

West Java Province presents a compelling case study for disaster-resilient EVCS planning due to its unique convergence of high population density (over 48 million inhabitants), significant economic importance (contributing 23% to national GDP), and substantial disaster vulnerability along the Pacific Ring of Fire [29]. The province faces multiple concurrent disaster risks including earthquakes, volcanic eruptions from 15 active volcanoes, landslides, and seasonal flooding, as demonstrated by recent events such as the devastating 2022 Cianjur earthquake that killed over 600 people and the 2024 West Java earthquake that damaged over 50.000 structures [30], [31], [32]. Indonesia, particularly West Java, is highly vulnerable to tsunamis due to its location along the Pacific Ring of Fire, with a historical record of major events that have caused severe damage. This vulnerability, combined with the region's high transportation demand, makes West Java an ideal setting for developing disaster-integrated EVCS planning methodologies that can subsequently be applied to other disaster-prone regions worldwide [33], [34].

This study addresses identified research gaps by developing and validating a comprehensive hybrid GIS-Fuzzy AHP system for identifying EVCS locations that are resilient to disaster risks while ensuring operational continuity during emergencies, specifically answering three critical questions: (1) How can disaster risk factors be systematically integrated into conventional EVCS planning criteria using Fuzzy AHP methodology? (2) What is the optimal weighting scheme for balancing routine accessibility with emergency preparedness in disaster-prone contexts? (3) How can the framework be adapted to other disaster-vulnerable regions in developing countries? The theoretical contribution lies in advancing the integration of disaster risk science with sustainable transportation planning by extending Fuzzy AHP's proven reliability in handling complex MCDM problems to disaster-resilient infrastructure planning, creating a comprehensive decision-support framework that ensures both routine operational efficiency and emergency preparedness capabilities essential for resilient urban mobility systems.

This study employs a hybrid GIS-Fuzzy AHP approach that combines spatial analysis capabilities with advanced multi-criteria decision-making techniques to address the complex challenges of disaster-resilient EVCS site selection. The methodological framework builds upon established MCDM practices while incorporating innovative disaster risk assessment components. The Fuzzy AHP component addresses uncertainty and imprecise information inherent in subjective decision-making processes, while the GIS component enables comprehensive spatial analysis of geographical, demographic, and infrastructure factors [35], [36]. The integrated approach allows for systematic evaluation of multiple criteria including accessibility, economic viability, technical feasibility, environmental impact, and critically, disaster risk vulnerability within a unified analytical framework. This methodology represents a significant advancement over conventional single-objective optimization methods by capturing intricate spatial relationships and multi-stakeholder requirements essential for resilient infrastructure planning in disaster-prone urban environments.

PERTANYAAN

1. Apa permasalahan global atau isu utama yang melatarbelakangi penelitian pada artikel tersebut? Jelaskan mengapa isu tersebut dianggap penting untuk diteliti.
2. Sebutkan tiga fokus penelitian terdahulu (*previous studies*) yang dijelaskan dalam introduction. Apa pendekatan atau metode yang digunakan pada penelitian sebelumnya?
3. Menurut Anda, apa kelemahan atau keterbatasan penelitian sebelumnya yang ingin diatasi oleh penulis?
4. Apa bentuk kebaruan (novelty) atau kontribusi utama yang ditawarkan artikel ini dibanding penelitian terdahulu?
5. Tuliskan tujuan penelitian (research objective) dari artikel tersebut, kemudian jelaskan apakah tujuan tersebut sudah sesuai dengan gap penelitian yang diidentifikasi sebelumnya.